1. Classification:

We've already seen classification predicts a category. Lets see the types of classification.

a)Binary classification:

It is a yes/no classification. Each input belongs to one of only two groups.

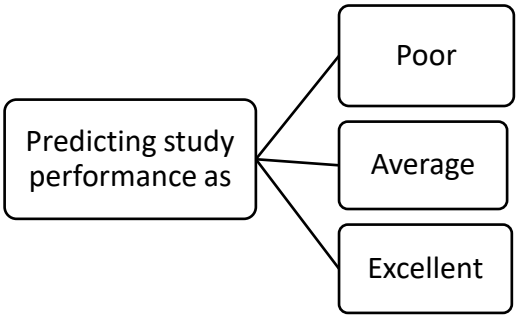
Example:

- Fraud / Not Fraud
- Pass / Fail
- Spam / Not Spam
- Disease / No Disease

b) Multi-class classification:

It is used when there are more than two possible categories. It predicts one out of **three or more** mutually exclusive classes (e.g., *Cat, Dog, or Bird*).

Example:



c) Multi-Label Classification

One item can belong to more than one category simultaneously. Unlike simple classification, which assigns only one label, this method allows multiple labels for a single input.

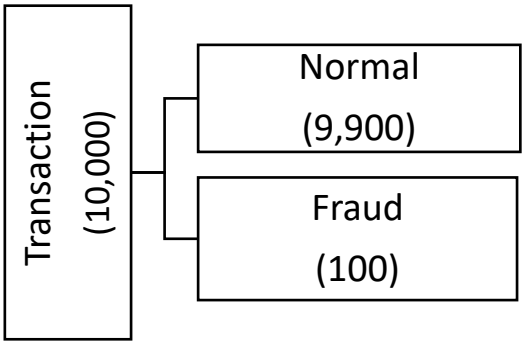
Example:

- a movie can be classified as both 'comedy' and 'romance'
- a photo can contain 'beach,' 'sunset,' and 'people.'

d) Imbalanced Classification

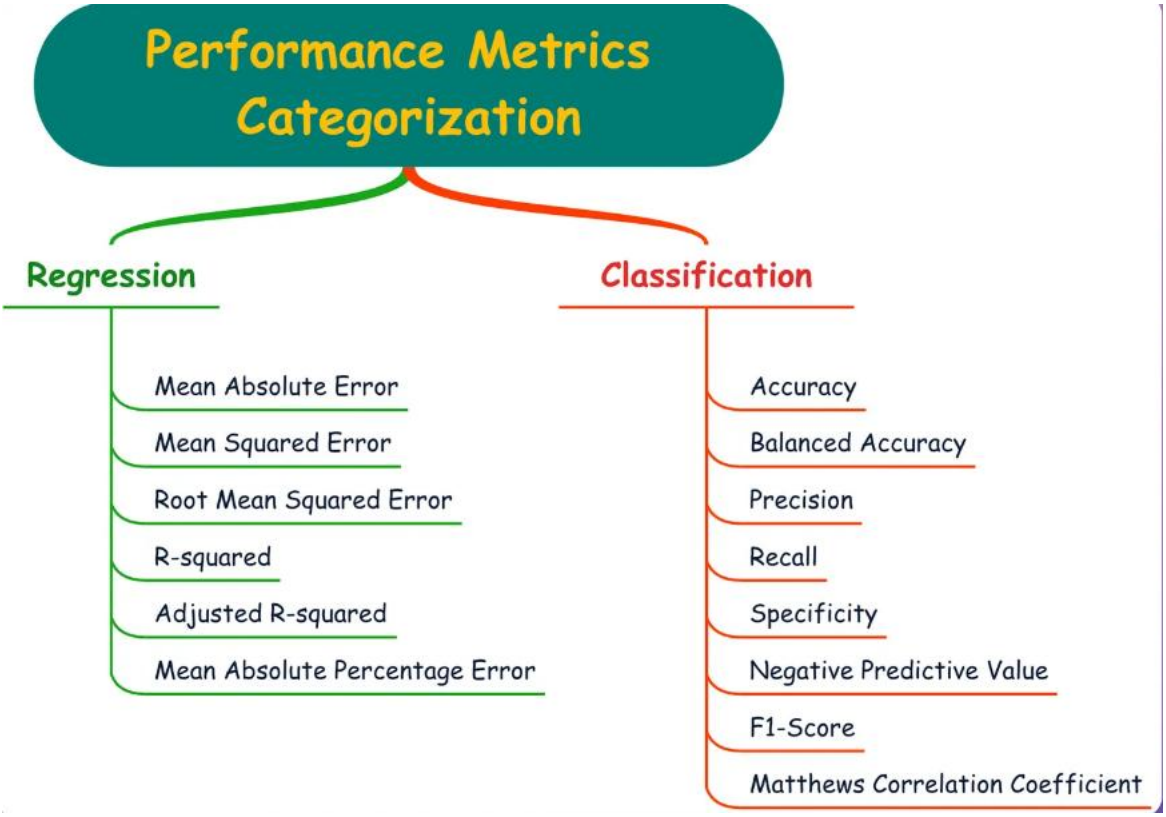
Imbalanced classification occurs when one data category has more examples than the other.

For example, in fraud detection



The normal transactions form the majority class, while fraud cases form the minority class. In such situations, a model may predict 'normal' every time and still show high accuracy, even though it fails to detect fraud.

To solve this, special techniques are used, and one among them is **evaluation metrics.**



Why isn't accuracy enough?

In the above case, accuracy = 9900/10000 → 99% accurate. It may sound excellent but model may fail to detect fraud transactions as normal transactions form majority class. Threfore, Accuracy can be misleading when classes are highly imbalanced.

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2. Confusion matrix:

Predicted Values \ Actual Values	Positive (1)	Negative (0)
Positive (1)	TP	FP
Negative (0)	FN	TN

a) **True Positive — TP:** The model predicts Positive and the actual result is also Positive.
b) **True Negative — TN:** The model predicts Negative.Reality is also Negative.
c) **False Positive — FP:** Model predicted positive when the actual result was negative.
d) **False Negative— FN:** Model predicted negative when the actual result was positive.

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3. Accuracy

measures how often the classifier correctly predicts. We can define accuracy as the ratio of the number of correct predictions and the total number of predictions.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

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4. Precision

It answers: When the model predicts Positive, how often is it actually correct?

Precision cares about False Positives.

$$Precision = \frac{TruePositive}{TruePositive + FalsePositive}$$

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5. Recall (Sensitivity)

It explains how many of the actual positive cases we were able to predict correctly with our model. False Negative is of higher concern than False Positive.

$$Recall = \frac{TruePositive}{TruePositive + FalseNegative}$$

6. F1 Score

When you need a single metric to balance both, use the F1 Score, which is the harmonic mean of precision and recall

$$F1 = 2. \frac{Precision \times Recall}{Precision + Recall}$$

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7. Problem Statement

Suppose a machine learning model is tested on a dataset of **100 patients** to predict whether they have a rare disease (Positive = Disease, Negative = No Disease).

From the model's test results, we get the following confusion matrix:

- **True Positives (TP) = 30** *(Model correctly predicted disease)*
- **False Positives (FP) = 20** *(Model wrongly predicted disease)*
- **True Negatives (TN) = 20** *(Model correctly predicted no disease)*
- **False Negatives (FN) = 10** *(Model missed the disease)*
- **Total Samples** = TP + FP + TN + FN = 30 + 20 + 20 + 10 = 100

Let's calculate the **Accuracy**, **Precision**, **Recall**, and **F1 Score**.

$$\text{Accuracy} = \frac{30 + 20}{30 + 20 + 20 + 10} = \frac{50}{80} = 0.625 \text{ (or } 62.5\%)$$

$$\text{Precision} = \frac{TP}{TP + FP} = \frac{30}{30 + 20} = \frac{30}{50} = 0.6 \text{ (or } 60\%)$$

$$\text{Recall} = \frac{TP}{TP + FN} = \frac{30}{30 + 10} = \frac{30}{40} = 0.75 \text{ (or } 75\%)$$

$$\text{F1 Score} = 2 \times \frac{0.6 \times 0.75}{0.6 + 0.75} = 2 \times \frac{0.45}{1.35} = 2 \times 0.3333 = 0.6667 \text{ (or } 66.67\%)$$

GitHub Repository — AI Essentials